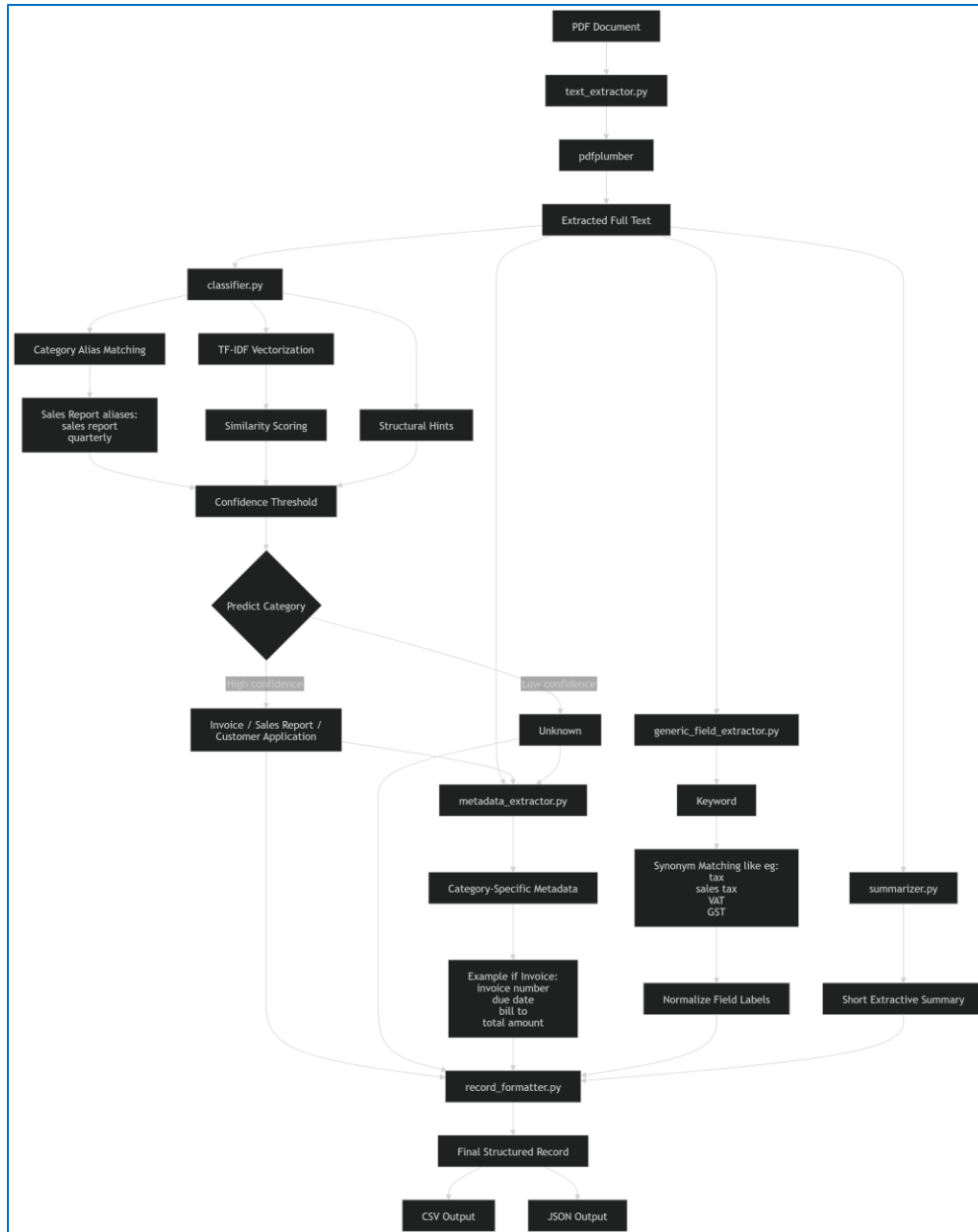


Week 03

Task 01

This is the local pipeline I created for the project.



Document categories were identified using TF-IDF, structural hints, and category aliases. Examples of category aliases include:

#### Invoice aliases:

- invoice no
- invoice number
- bill to
- billed to
- amount due

#### Sales Report aliases:

- quarterly
- monthly
- annual
- Q1 / Q2
- executive summary
- revenue breakdown

#### Customer Application aliases:

- account application
- application form
- application id
- applicant name
- full name
- date of birth

To make the pipeline more robust, I created a synonym list based on the prototype list used for TF-IDF. This helped maximize keyword extraction and reduce missed matches. An example of the synonym list is:

Tax:

- sales tax
- VAT
- GST
- tariff
- levy
- duty

Total amount due:

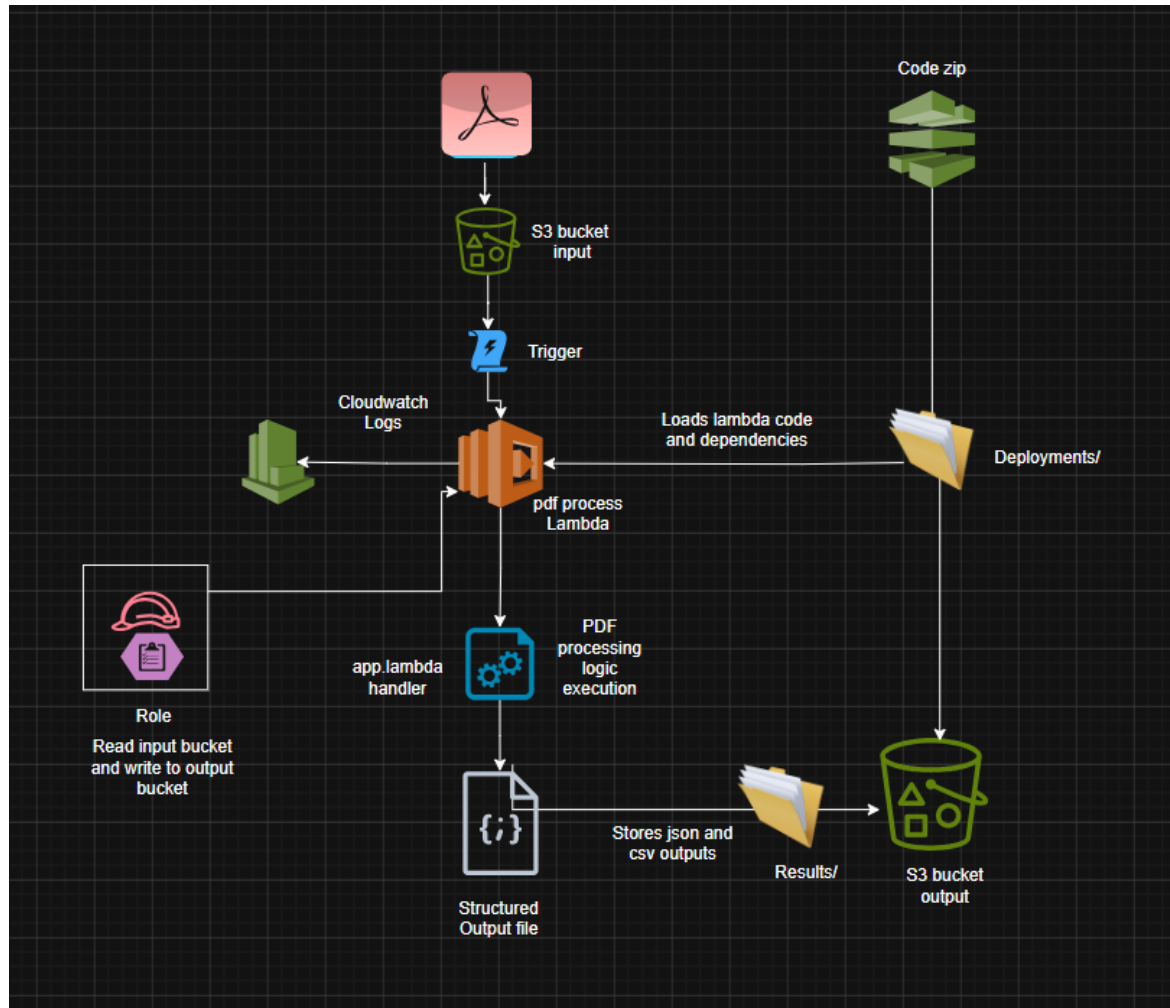
- total amount due
- amount due
- balance due
- amount payable
- total payable
- grand total

Then I tested it with sample pdfs. They were synthetically generated. For some pdfs headers were also removed. Yet the results were quite good.

```
✓ TASK 01      files > pdf_pipeline_aws > local_results > {} test.json > ...
└─ files\pdf_pipeline_aws 1 {
  > build 2
  > docs 3
  └─ lambda_src 4
    > __pycache__ 5
    app.py 6
    classifier.py 7
    generic_field_extrac... 8
    metadata_extractor... 9
    record_formatter.py 10
    requirements.txt 11
    summarizer.py 12
    text_extractor.py 13
  └─ local_results 14
    invoice_001.csv 15
    {} invoice_001.json 16
    test.csv 17
    {} test.json 18
  > sample_pdfs 19
  > scripts 20
  .gitignore 21
  deploy.sh 22
  README.md 23
  ! template.yaml 24
  25 {
  26   "document_id": "b3125b69-8079-4076-adf4-2a364dd25dc5",
  27   "filename": "test.pdf",
  28   "source_path": "D:\\\\internship_aws\\\\week_03\\\\task_01\\\\files\\\\pdf_pipeline_aws\\\\test.pdf",
  29   "category": "Invoice",
  30   "best_category_match": "Invoice",
  31   "confidence": 0.8367,
  32   "category_scores": {
  33     "Invoice": 1.0,
  34     "Sales Report": 0.08,
  35     "Customer Application": 0.1633
  36   },
  37   "category_alias_scores": {
  38     "Invoice": 14.0,
  39     "Sales Report": 2.0,
  40     "Customer Application": 1.0
  41   },
  42   "category_tfidf_scores": {
  43     "Invoice": 0.3008,
  44     "Sales Report": 0.0196,
  45     "Customer Application": 0.1158
  46   },
  47   "category_structural_scores": {
  48     "Invoice": 6.1,
  49     "Sales Report": 0.0,
  50     "Customer Application": 0.0
  51   },
  52   "num_pages": 1,
  53   "char_count": 843,
  54   "extractor": "pdfplumber",
  55   "summary": "Invoice No: INV-001 Invoice Date: 13/05/2026 Due Date: 20/05/2026 From: Bright Star Traders"
```

```
,
  "num_pages": 1,
  "char_count": 843,
  "extractor": "pdfplumber",
  "summary": "Invoice No: INV-001 Invoice Date: 13/05/2026 Due Date: 20/05/2026 From: Bright Star Traders",
  "metadata": {
    "invoice_number": "INV-001",
    "invoice_date": "13/05/2026",
    "due_date": "20/05/2026",
    "bill_to": "Bright Star Traders",
    "customer_email": "info@abcdigitalsolutions.com",
    "subtotal": null,
    "tax": null,
    "total_amount_due": null,
    "currency": null
  },
  "keyword_fields": {
    "invoice_number": "INV-001",
    "invoice_date": "13/05/2026",
    "due_date": "20/05/2026",
    "subtotal": 53000.0,
    "tax": 0.0,
    "total_amount_due": 50000.0,
    "customer_name": "Bright Star Traders",
    "email": [
      "info@abcdigitalsolutions.com",
      "accounts@brightstartraders.com"
    ],
    "phone": [
```

After making sure that the pipeline works accordingly and all the test pdfs passed. I moved onto the AWS pipeline.



## Uploading pdfs to s3 bucket

Files and folders

Configuration

Files and folders (7 total, 185.8 KB)

Find by name

| Name                                         | Folder | Type            | Size     | Status      | Error |
|----------------------------------------------|--------|-----------------|----------|-------------|-------|
| <a href="#">customer_application_001.pdf</a> | -      | application/pdf | 2.6 KB   | ✓ Succeeded | -     |
| <a href="#">customer_application_002.pdf</a> | -      | application/pdf | 2.6 KB   | ✓ Succeeded | -     |
| <a href="#">invoice_001.pdf</a>              | -      | application/pdf | 2.5 KB   | ✓ Succeeded | -     |
| <a href="#">invoice_002.pdf</a>              | -      | application/pdf | 2.4 KB   | ✓ Succeeded | -     |
| <a href="#">sales_report_q2_2025.pdf</a>     | -      | application/pdf | 2.8 KB   | ✓ Succeeded | -     |
| <a href="#">sales_report_q3_2025.pdf</a>     | -      | application/pdf | 2.9 KB   | ✓ Succeeded | -     |
| <a href="#">test.pdf</a>                     | -      | application/pdf | 170.0 KB | ✓ Succeeded | -     |

After uploading, the lambda function was executed, yielding the following results.

Amazon S3 > Buckets > pdf-pipeline-results-eman > results/ > json/

Objects Properties

Objects (7)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

| <input type="checkbox"/> | Name                                                      | Type | Last modified                      | Size   | Storage class |
|--------------------------|-----------------------------------------------------------|------|------------------------------------|--------|---------------|
| <input type="checkbox"/> | <a href="#">17210100-1948-48f7-bd83-9f45fa30082d.json</a> | json | May 14, 2026, 18:40:59 (UTC+05:00) | 3.2 KB | Standard      |
| <input type="checkbox"/> | <a href="#">1b2d884e-75b0-4869-905a-140068d26bba.json</a> | json | May 14, 2026, 18:41:01 (UTC+05:00) | 2.1 KB | Standard      |
| <input type="checkbox"/> | <a href="#">924dff4d-4155-4185-87df-6464639a13fb.json</a> | json | May 14, 2026, 18:40:58 (UTC+05:00) | 4.1 KB | Standard      |
| <input type="checkbox"/> | <a href="#">a67eb17c-7d4d-40fa-a6f5-aab3fd04a7d6.json</a> | json | May 14, 2026, 18:40:59 (UTC+05:00) | 3.2 KB | Standard      |
| <input type="checkbox"/> | <a href="#">c5c89bcd-4c93-4d52-8c02-8ce08c67f3cc.json</a> | json | May 14, 2026, 18:41:01 (UTC+05:00) | 4.1 KB | Standard      |
| <input type="checkbox"/> | <a href="#">f397af41-6a6e-46a8-a4f9-1726b7f726f4.json</a> | json | May 14, 2026, 18:41:00 (UTC+05:00) | 2.1 KB | Standard      |
| <input type="checkbox"/> | <a href="#">fe6ed039-eb8e-4967-a590-964e292c3484.json</a> | json | May 14, 2026, 18:40:58 (UTC+05:00) | 4.1 KB | Standard      |

The pipeline generates both CSV and JSON outputs. The JSON output preserves the complete structured representation of each processed document, making it easier to inspect the full metadata, category, generic fields, and summary together. The CSV output provides a flat tabular format, which is more suitable for spreadsheet analysis, reporting, and importing into relational databases.

```
pdf-pipeline-results-eman.s3.us-east-1.amazonaws.com/results/json/c5c89bcd-4c93-4d52-8c02-8ce08c67f3cc.json?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Content-Sha256=UNSIGNED-PAL...
retty-print

{"document_id": "c5c89bcd-4c93-4d52-8c02-8ce08c67f3cc",
 "filename": "test.pdf",
 "source_bucket": "pdf-pipeline-input-eman",
 "source_key": "test.pdf",
 "s3_uri": "s3://pdf-pipeline-input-eman/test.pdf",
 "category": "Invoice",
 "best_category_match": "Invoice",
 "confidence": 0.8367,
 "category_scores": {
   "Invoice": 1,
   "Sales Report": 0.08,
   "Customer Application": 0.1633
 },
 "category_alias_scores": {
   "Invoice": 14,
   "Sales Report": 2,
   "Customer Application": 1
 },
 "category_tfidf_scores": {
   "Invoice": 0.3008,
   "Sales Report": 0.0196,
   "Customer Application": 0.1158
 },
 "category_structural_scores": {
   "Invoice": 6.1,
   "Sales Report": 0,
   "Customer Application": 0
 },
 "num_pages": 1,
 "char_count": 843,
 "extractor": "pdfplumber",
 "summary": "Invoice No: INV-001 Invoice Date: 13/05/2026 Due Date: 20/05/2026 From: ABC Digital Solutions Office No. 12, Business Plaza, Islamabad, Pakistan Phone: +92 300 1234567 Email: info@abcdigitalsolutions.com Bill To: Bright Star Traders Main Boulevard, Lahore, Pakistan Phone: +92 321 9876543 Email: accounts@brightstartraders.com Description Quantity Unit Price Total Website Design service 1 PKR 35,000 PKR 35,000 Monthly Website Maintenance 1 PKR 10,000 PKR 10,000 Domain & Hosting Setup 1 PKR 8,000 PKR 8,000 Subtotal: PKR 53,000 Tax: PKR 0 Discount: PKR 3,000 Grand Total: PKR 50,000 Payment Method: Bank Transfer Payment Details: Account Title: ABC Digital Solutions Bank Name: HBL Account/IDAN: PK00HARB0000000001234567 Notes: Thank you for your business. Please make the payment by the due date. Authorized Signature: ABC Digital Solutions",
 "metadata": {
   "invoice_number": "INV-001",
   "invoice_date": "13/05/2026",
   "due_date": "20/05/2026",
   "bill_to": "Bright Star Traders",
   "customer_email": "info@abcdigitalsolutions.com",

```

```
{
  "document_id": "9176319b-633b-41f0-887c-5c75986564d1",
  "filename": "invoice_001.pdf",
  "source_path": "D:\\internship_aws\\week_03\\task_01\\files\\pdf_pipeline_aws\\sample_pdfs\\invoice_001.pdf",
  "category": "Invoice",
  "confidence": 0.9331,
  "category_scores": {
    "Invoice": 1.0,
    "Sales Report": 0.0669,
    "Customer Application": 0.0309
  },
  "alias_scores": {
    "Invoice": 18.5,
    "Sales Report": 2.0,
    "Customer Application": 0.0
  },
  "tfidf_scores": {
    "Invoice": 0.6241,
    "Sales Report": 0.0423,
    "Customer Application": 0.0552
  },
  "structural_scores": {
    "Invoice": 9.1,
    "Sales Report": 0.0,
    "Customer Application": 0.0
  },
  "num_pages": 1,
  "char_count": 509,
  "extractor": "pdfplumber",
  "summary": "INVOICE Invoice Number: INV-2025-0042 Invoice Date: 2025-09-15 Due Date: 2025-09-15 Bill To: Acme Corporation 123 Business Park, Karachi, Pakistan Email: billing@acme-corp.com Description Qty Unit Price Total Cloud Hosting (Monthly) 1 $450.00 $450.00 Premium Support Plan 1 $200.00 $200.00 Data Backup Storage (1B) 3 $75.00 $225.00 Subtotal: $875.00 Tax (17%): $148.75 Total Amount Due: $1023.75 Payment Terms: Net 30 days. Please remit payment by the due date to avoid late fees. Thank you for your business.",
  "metadata": {
    "invoice_number": "INV-2025-0042",
    "invoice_date": "2025-09-15",
    "due_date": "2025-09-15",
    "bill_to": "Acme Corporation",
    "customer_email": "billing@acme-corp.com",
    "subtotal": 875.0,
    "tax": 148.75,
  }
}
```